

Management of patients with neurological diseases

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الحديث الجانبي يشتت الانتباه



من فضلك اغلق الهاتف



Epilepsy

3- Epilepsy

- Definition
- Causes / risk factors
- Stages / phases
- Manifestations
- Types
- Diagnosis
- Managements

Definition: chronic neurological disorder characterized by **recurrent, unprovoked** seizures due to **abnormal electrical activity** in the brain.

A person is generally diagnosed with epilepsy after experiencing **two or more** unprovoked **seizures** at least **24 hours** apart.



Causes / Risk Factors:



Children	Birth injury, genetic syndromes, infections, developmental disorders
Adults	Trauma, brain tumors, stroke, infections
Older Adults	Stroke, Alzheimer's, brain atrophy

Other risk factors:

- **Family history of epilepsy**
- **High fever (febrile seizures in children)**
- **Sleep deprivation**
- **Substance abuse or withdrawal**
- **Brain infections (e.g., meningitis, encephalitis)**

CAUSES AND RISK FACTORS FOR EPILEPSY



Brain Injury



Genetics &
Family History



Abnormal Brain Development
in Womb or Infancy



Autoimmunity



High Fevers



Certain Medications



Infectious
Diseases



Anxiety & Stress



Electrolyte &
Nutrient Imbalances

Stages of seizure

Prodrome ,(Aura)

Ictal (tonic , clonic)

Postictal



1- Prodromal

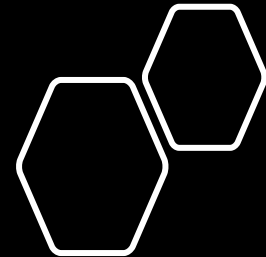
Before seizure

Occur several hours or even days before the actual seizure

Confusion, anxiety, irritability, headache, tremor, anger, and mood disturbances



**An "Aura" is
a symptom
that can
occur before
a seizure**



- **Aura is the earliest sign of seizure activity and the beginning of the ictal phase**

- Acidic taste, offensive smells
- **Déjà Vu** (feeling of **familiarity** with a person, place, or thing without having experienced it)
- **Jamais vu** (feeling of **unfamiliarity** with a person, place, or thing despite having already experienced it)
- Dizziness
- Flickering vision, Vision loss or blurring
- Hallucinations
- Head, arm, or leg pain
- Nausea/stomachache
- Numbness
- Ringing or buzzing sounds
- Tingling



2- Ictal

Attack of seizure

Tonic

Arm or leg stiffening
Loss of bladder and/or
bowel control
loss of balance
Difficulty breathing
Sweating

Clonic

Repeated, rapid muscle jerks
Rhythmic shaking of arms,
legs, or face
Involuntary body movements
loss of consciousness
Frothy saliva

Tonic-clonic (grand mal) seizure



Tonic phase
(10 to 30 seconds)



Clonic phase
(30 to 60 seconds)



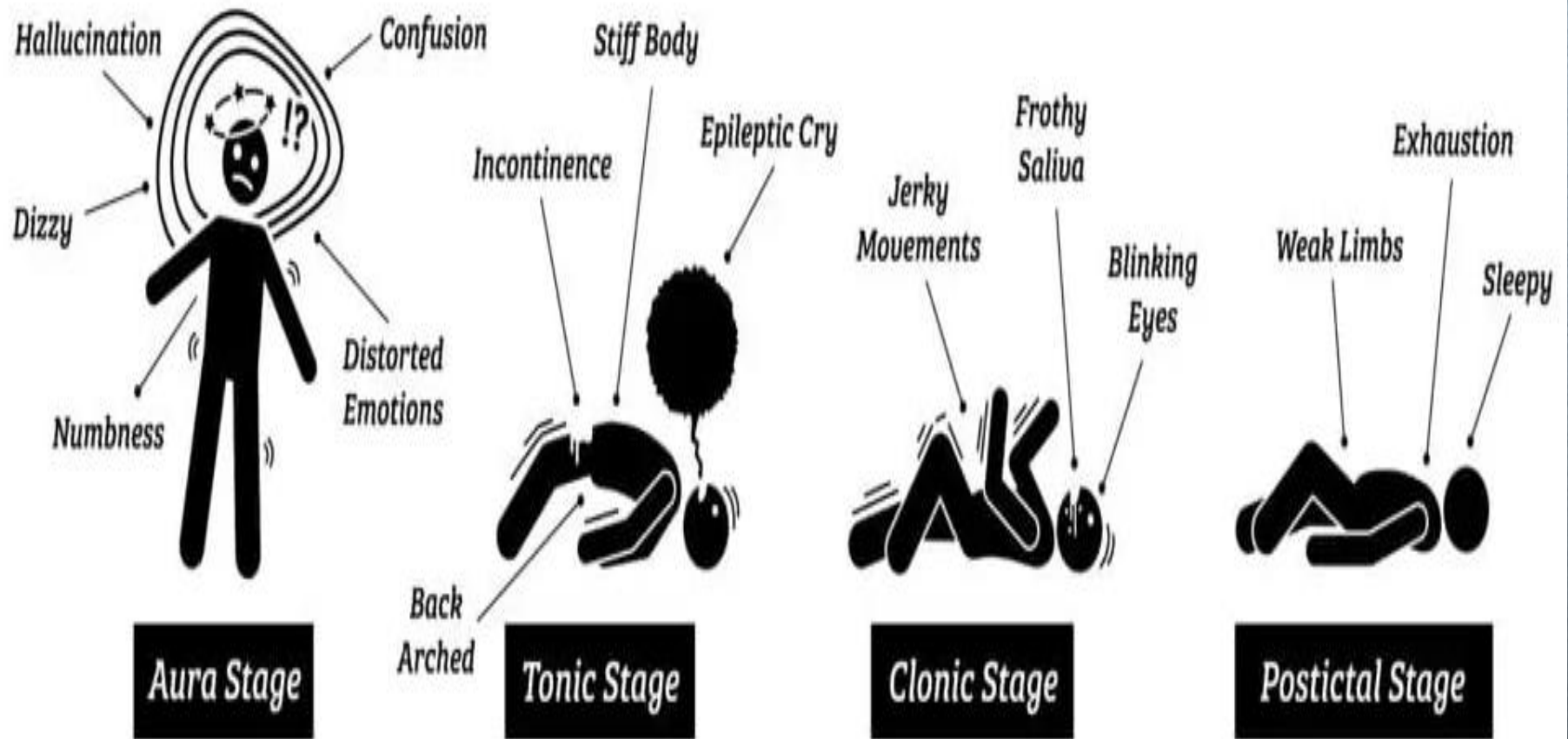
3- Postictal

Recovery period after seizure

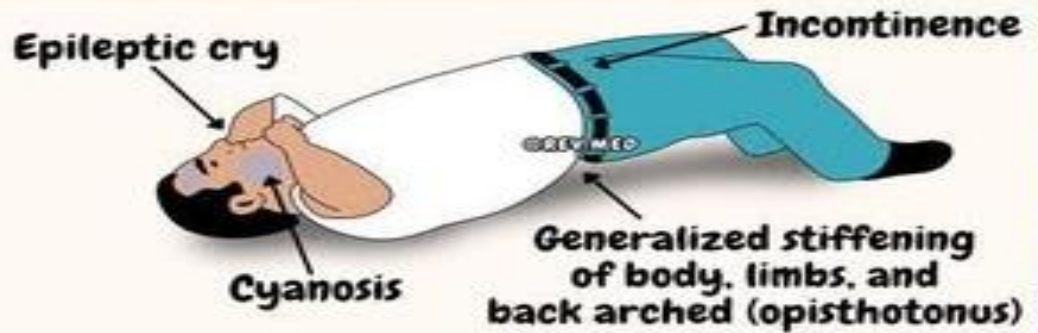
Some people recover immediately, while others may require minutes, hours to feel like they're back at their baseline

Arm or leg weakness
Confusion
Drowsiness & Memory loss
Thirst
General malaise, sleep
Embarrassment
Headaches/migraine

Stages of a Seizure



Tonic Phase



Clonic Phase



Postictal slowing



Phases of Seizures

Phases of seizures describe the sequence of events that may occur before, during, and after a seizure. Not all seizures have all phases.



Summary		
Phase	Timing	Main Features
Prodromal Phase	Hours-days Before	Mood changes, fatigue
Aura	Seconds-to Minutes Before	Sensory warning
Ictal	During Seizure	Active seizure
Postictal Phase	Minutes-Hours After	Confusion, fatigue



TYPES

Focal/Partial



Simple (pt is aware)



Complex (impaired awareness)

Generalized



Tonic



A tonic

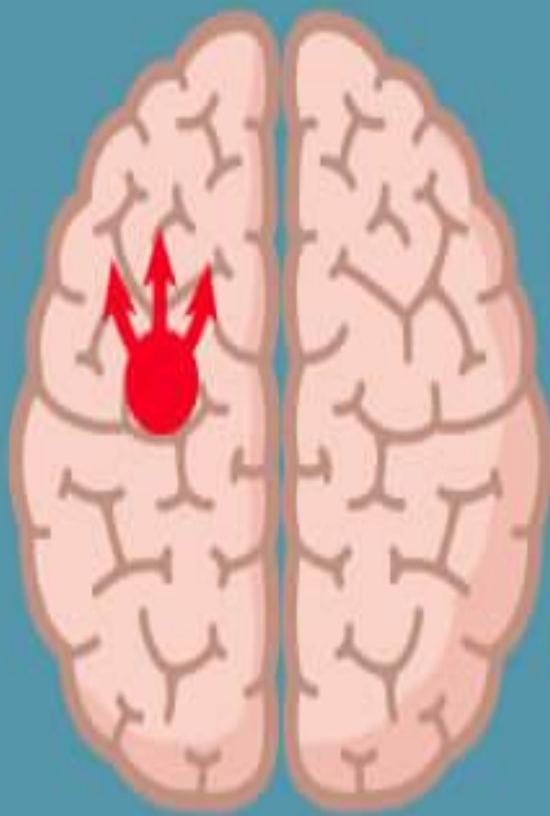


Tonic clonic

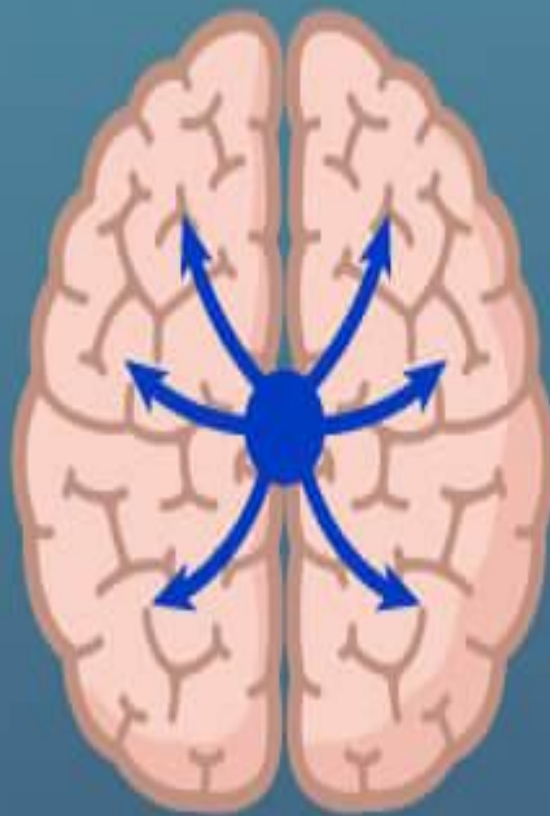


Myoclonic

Focal Onset



Generalized Onset



FOCAL SEIZURE

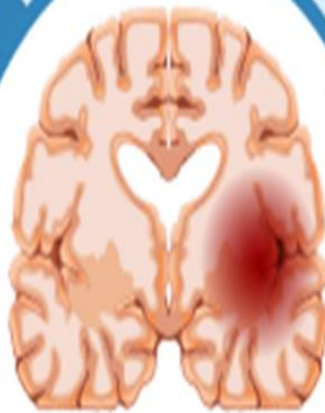


GENERALIZED SEIZURE





Healthy



Focal seizure



**Generalized
epilepsy**

symptoms of focal seizures

1) Simple focal

 **Consciousness is preserved**

Symptoms:

- Involuntary movements in one part of the body (arm, leg, or face)
- Tingling or numbness
- Strange sensations (unusual smell or taste)
- Sudden emotions (fear, anxiety)
- The person remembers everything during the seizure

2) Complex focal

 **Consciousness is affected**

Symptoms:

- Staring or blanking out
- No response to speech or surroundings
- Repetitive automatic movements (lip smacking, hand rubbing, chewing)
- Confusion
- Usually no memory of the event

symptoms of generalized seizures

Tonic seizure

- **Sudden stiffening of the body**
- No rhythmic jerking
- Often causes falls

Atonic seizure

- **Sudden loss of muscle tone**
- Body goes limp (“drop attack”)
- Can cause falls and injuries

Tonic-clonic seizure

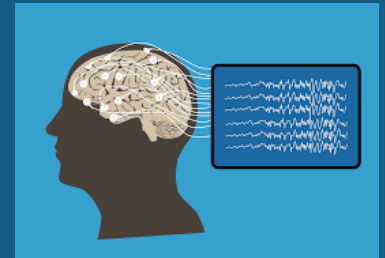
- **Tonic phase:**
- Sudden stiffening of the whole body
- Loss of consciousness
- Falling down
- **Clonic phase:**
- Repeated jerking movements of the body
- Rhythmic muscle contractions

Myoclonic seizure

- Sudden, brief muscle jerks
- Usually affects arms or upper body
- Very short (seconds)

Assessment and diagnostic studies :

- ❖ Detailed history of **seizure**: onset, duration, triggers, symptoms
- ❖ **Neurological examination**
- ❖ **EEG**: detects abnormal brain activity
- ❖ **MRI/CT scan**: identify presence of lesions ,tumor or stroke
- ❖ **Blood tests**: metabolic or infectious causes
- ❖ **Lumbar puncture**: if CNS infection is suspected



COMMON SEIZURE TRIGGERS



Increased
Physical or
Emotional Stress



Fatigue,
Lack of sleep,
or Poor Sleep



Anxiety or Panic



Alcohol or Drug
Use or Withdrawal
Symptoms



Changing or
Skipping Medications



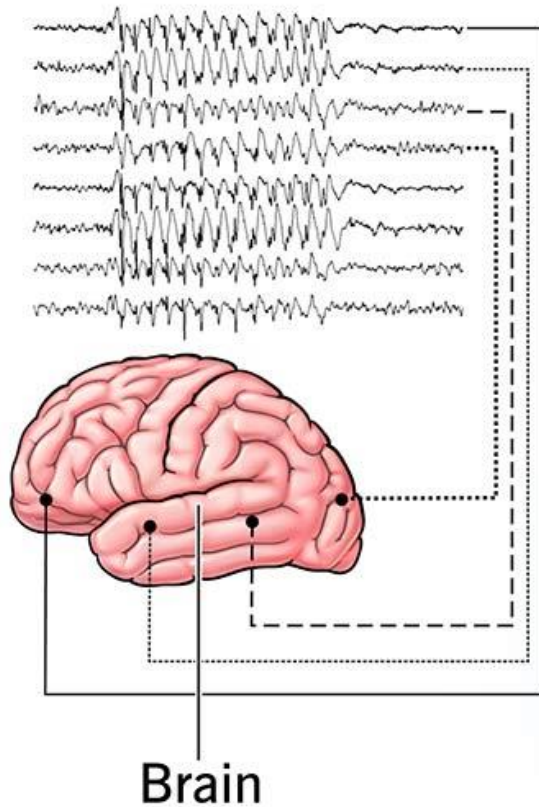
Overstimulation
by Loud Noises,
Lights, and
Electronics



Hormone Changes
(Puberty, Menstrual
Cycle, Pregnancy or
Menopause)

Electroencephalogram (EEG)

EEG (scan of brainwaves)



Electrodes
glued to scalp



DIAGNOSING EPILEPSY

The process of diagnosing epilepsy utilizes your medical history, exam, blood work and advanced diagnostic testing.



Medical
History



Physical
Exam



Blood Chemistry
Study



Electroencephalogram
(EEG)



CT Scan, MRI
and/or PET Scans

DRJOCKERS.COM
SUPERCHARGE YOUR HEALTH

Drug of choice for status epilepticus?

A. Phenytoin



B. Diazepam



C. Valproic acid



D. Carbamazepine



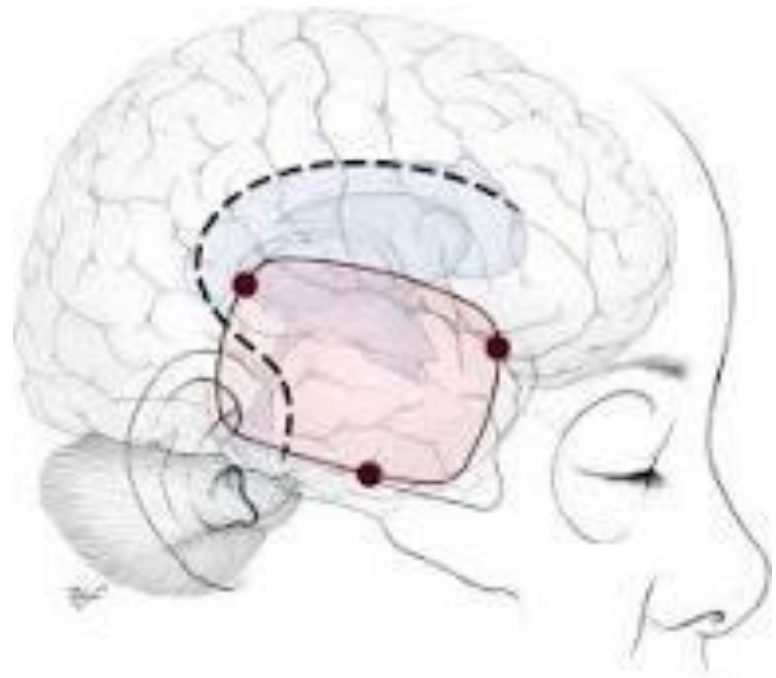
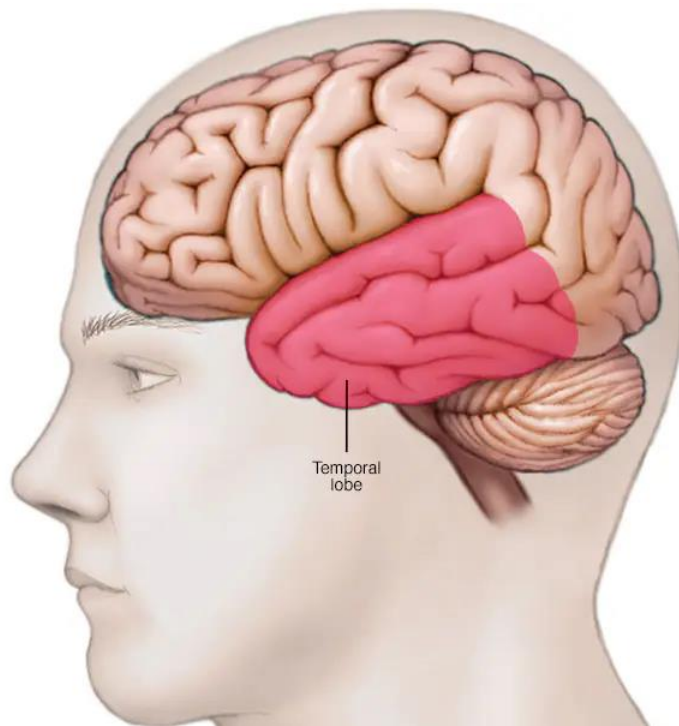
Management

- Medications (Anti-Epileptic Drugs - AEDs)
- Selection depends on seizure type, age, comorbidities
- Monitor side effects:
- Dizziness, rash, sedation, hepatotoxicity, bone marrow suppression.

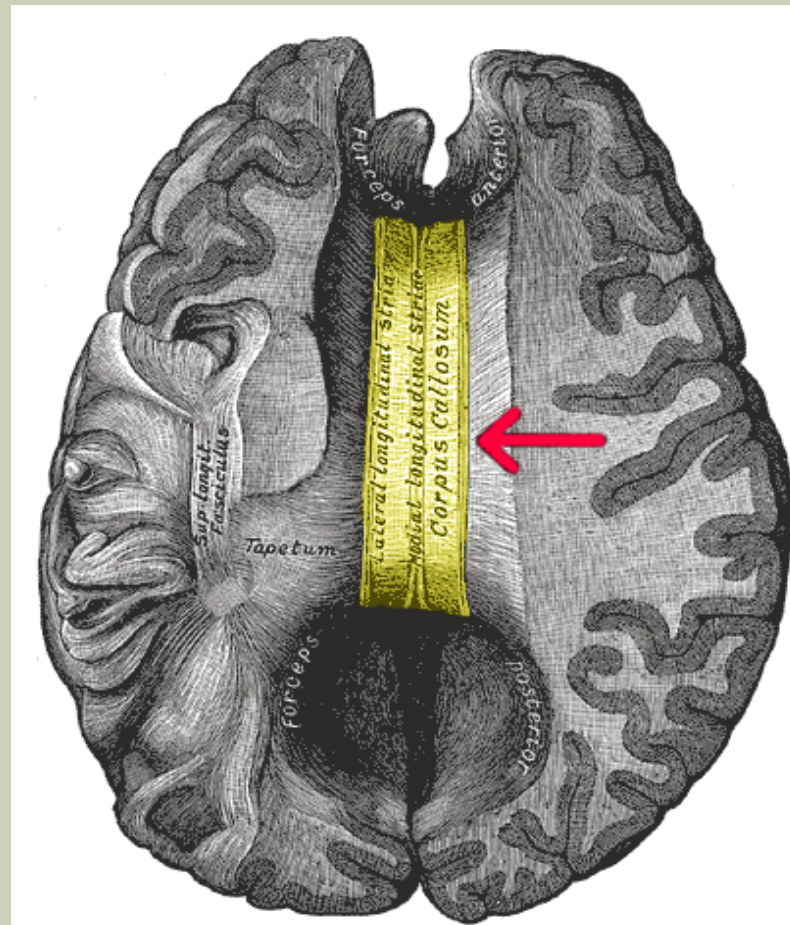
Surgical Options

- For refractory epilepsy (not controlled by medications)
 - **Temporal lobectomy**
 - **Corpus callosotomy**
 - **Vagus nerve stimulation (VNS)**
 - **Responsive neurostimulation (RNS)**

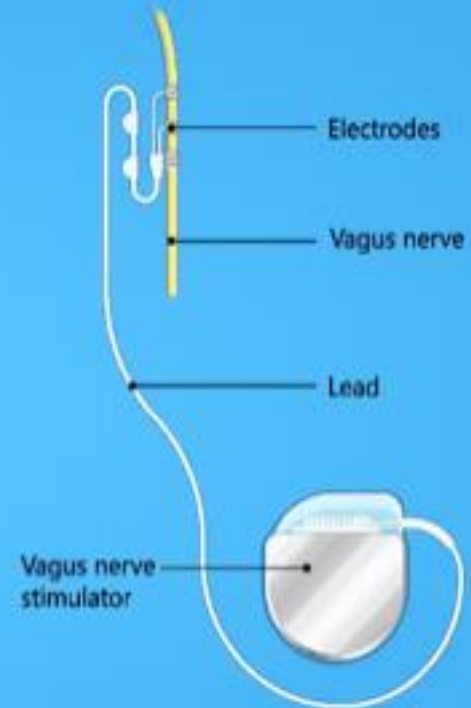
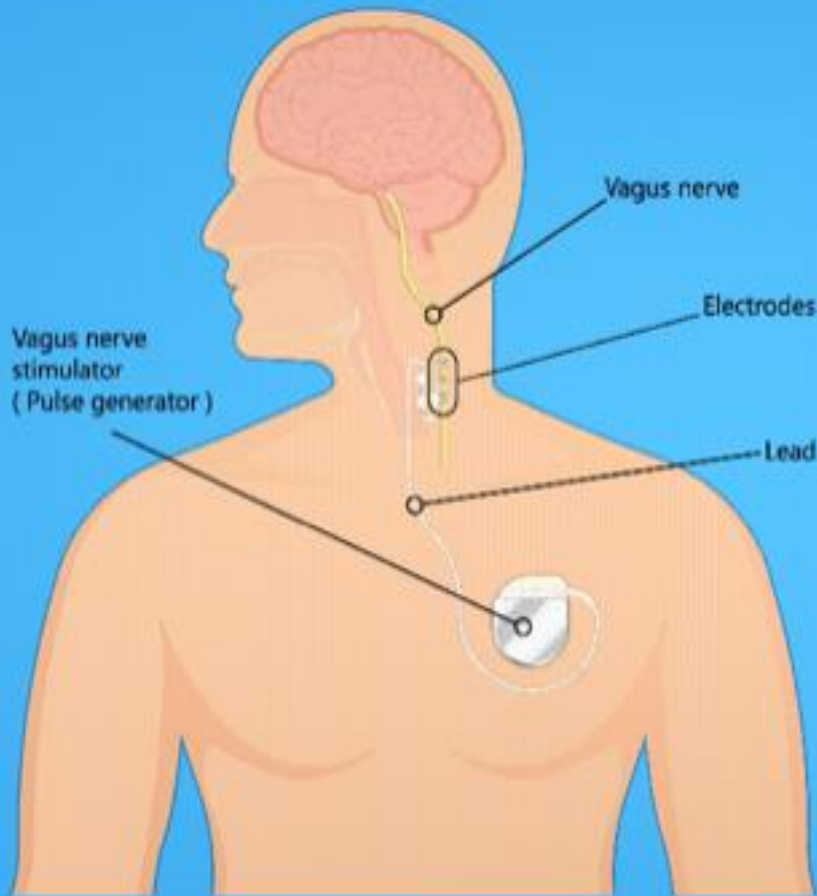
Temporal lobectomy



Corpus callosotomy



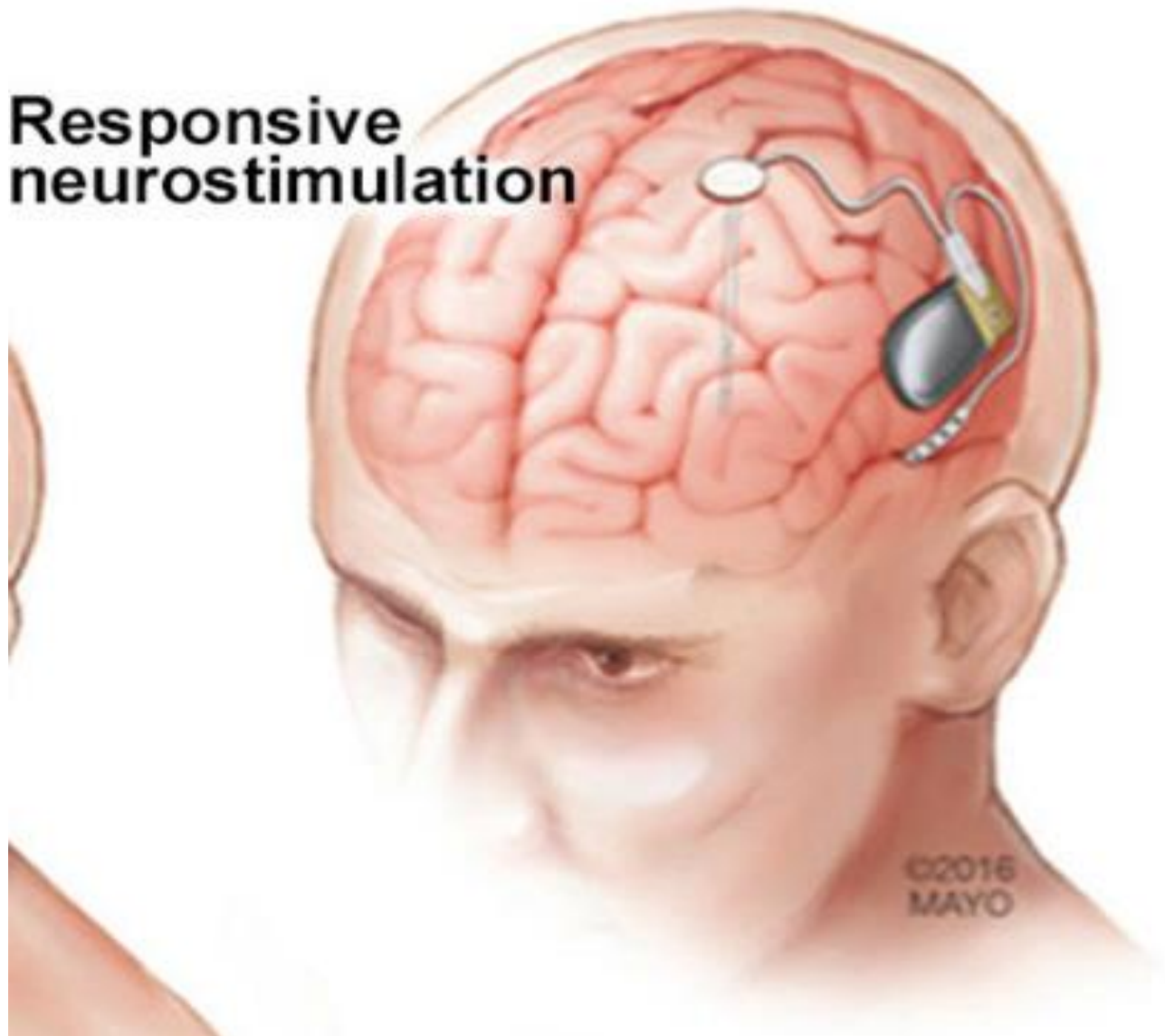
Vagus nerve stimulation (VNS)



Vagus Nerve Stimulation
(VNS)

(VNS)
Vagus Nerve Stimulation

Responsive neurostimulation



SEIZURE FIRST AID



Stay with the person
and start timing
the seizure



Ensure safety by
removing potentially
harmful objects



Turn the person
onto their side
if unconscious



Loosen clothing
around the neck



Do not put
anything in the
person's mouth



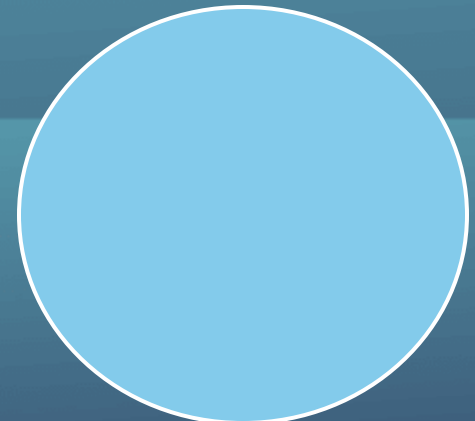
Do not restrain
the person



Put something
soft under the head



Stay with the
person until the
seizure subsides



Nursing management :

Pre-ictal phase (Prodromal / Aura phase)

-  Before the seizure or early warning signs
- Observe for **early warning signs** (aura, behavior changes, headache, anxiety)
- Ensure patient **safety** (prepare environment)
- Keep the patient in a **safe area**
- **Raise** bed rails
- Remove **sharp** or **dangerous** objects
- **Reduce** stress and provide **reassurance**
- Ensure **antiepileptic** medications are taken on time
- **Document** any aura or pre-seizure symptoms

Nursing management :

Ictal phase

👉 During the seizure

- Turn the patient to a **lateral** (side-lying) position if possible
- **Protect** the head with padding
- Remove nearby **harmful objects**
- **Loosen** tight clothing, especially around the neck
- **Do NOT put anything in the mouth**
- **Do NOT try to stop the movements**
- **Do NOT restrain**
- Monitor:
 - Duration of seizure
 - Type of movements
 - Breathing and level of consciousness
- Provide **oxygen** if needed
- Prepare **emergency medications** if seizure is prolonged (e.g., benzodiazepines as prescribed)

Nursing management :

Postictal phase (After seizure)

👉 Recovery phase after the seizure

- Maintain **airway** patency
- Keep patient in **side-lying** position if drowsy
- Monitor **vital signs** (BP, pulse, oxygen saturation)
- Assess level of **consciousness** gradually
- Provide a **calm** and **quiet** environment
- Manage symptoms:
 - Fatigue
 - Headache
 - Confusion
- **Document** full seizure details:
 - Time of onset and duration
 - Type of seizure
 - Patient response
- Provide **emotional support** after recovery



Summary



- **List 3 risk factors of seizure**
- **List 4 manifestations of ictal phase of seizure .**
- **Differentiate between focal and generalized seizure**



Thank
you